

Nitish Shukla

PhD Student in Computer Science

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Professional summary

Second year PhD student specializing in biometric security and LLM preference optimization, with a strong academic foundation in mathematics and balanced experience across industry and academia. Pioneered state-of-the-art methods in Face Demorphing and morph attack detection, developing unified pipelines across biometric modalities for high reliability and real-world usability. Skilled in GANs, diffusion models, LLM, and transformers, with a focus on advancing secure, robust biometric technology.

Links

Portfolio:

nitishshukla86.github.io,

LinkedIn: [linkedin.com](https://www.linkedin.com/in/nitishshukla86),

Google Scholar:

[scholar.google.com](https://scholar.google.com/citations?user=NitishShukla86).

Skills

Face Biometrics (*Experienced*),

Deep Learning (*Experienced*),

Computer Vision
(*Experienced*),

Adversarial ML (*Skillful*),

Bayesian Theory,

Graph Theory,

Mathematics,

PyTorch,

Scikit-learn,

OpenCV,

MLFlow,

Pandas,

LLM Planning (*Skillful*).

References

Arun Ross, Michigan State University
(rossarun@msu.edu),

K.V. Subramanayam, CMI
(kv@cmi.ac.in, +91 9445 390 492),

Sudipta Banerjee, IIIT-H
(sudipta.b@iiit.ac.in).

Employment history

PhD Intern, May 2025 - Aug 2025

Adobe Research

Validating style images using LLM induced attributes.

PhD Intern, May 2024 - Aug 2024

Adobe Research

- Investigated effects of mutable text embedding in text-to-image generation.
- Planning via LLMs: Employed LLMs to generate plans executing various simulated tasks.

Data Scientist II, Aug 2022 - Aug 2023

Micron Technology

R&D Engineer, May 2019 - Aug 2020

Next Education

Research Publications

•**Nitish Shukla**, Arun Ross, “diffDemorph: Extending Reference-Free Demorphing to Unseen Faces”, In Proceedings of IEEE International Conference on Image Processing (ICIP), 2025.

•**Nitish Shukla**, Arun Ross, “dc-GAN: Dual-Conditioned GAN for Face Demorphing From a Single Morph”, In Proceedings of IEEE International Conference on Automatic Face and Gesture Recognition (FG), 2025.

•**Nitish Shukla**, Arun Ross, “Metric for Evaluating Performance of Reference-Free Demorphing Methods”, In Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), 2025.

•**Nitish Shukla**, Arun Ross, “Face Demorphing via Identity Preserving Image Decomposition”, In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2024.

•**Nitish Shukla**, Sudipta Banerjee, “Generating Adversarial Attacks in Latent Space”, In Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023.

•**Nitish Shukla**, “SDeMorph : Towards Better Component Face Extraction from Single Morph”, In Proceedings of IEEE International Joint Conference on Biometrics (IJCB), 2023

•**Nitish Shukla**, Arun Ross, “Privacy Preserving Facial Demorphing”, Under Review, 2024.

Education

PhD in Computer Science, Jan 2024 - Present

Michigan State University, MI, USA

Working on single image reference free demorphing methods with Prof. Arun Ross.

Master of Science in Computer Science, Dec 2020 - Jun 2022

Chennai Mathematical Institute, Chennai, India

Master of Science in Mathematics, Jul 2017 - Jun 2019

Indian Institute of Technology (IIT), Guwahati, Guwahati, India